SAECOIN

USER INTERFACE SPECIFICATION

Project: SAECOIN
Ticker: SAC (Provisional)
Document: UI_Spec.md
Version: 1.0
Status: Planning
Parent Documents: PRD.md, AppFlow.md

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1. DOCUMENT PURPOSE

This document defines the visual design system, user interface components, layouts, navigation, screen structures, and interaction behavior of the SAECOIN software ecosystem.

All SAECOIN applications must maintain a consistent visual identity.

The design must prioritize:

- Simplicity.
- Professional Appearance.
- Readability.
- Security Awareness.
- Accessibility.
- Performance.
- Responsive Layouts.

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2. BRAND IDENTITY

2.1 Project Identity

Project Name: SAECOIN

Coin Name: SAECOIN

Ticker Symbol: SAC (Provisional)

Brand Personality:

- Modern.
- Technological.
- Trustworthy.
- Decentralized.

- Secure.
- International.

2.2 Logo Requirements

SAECOIN must have a recognizable cryptocurrency logo.

Logo requirements:

- Original design.
- Simple geometric identity.
- Suitable for mobile applications.
- Suitable for desktop applications.
- Suitable for website headers.
- Suitable for small favicon displays.
- Support transparent backgrounds.

The logo must work on both dark and light backgrounds.

Final logo artwork will be created separately.

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3. DESIGN SYSTEM

3.1 Color Palette

The initial design direction uses a Dark Technology Theme.

Primary Colors

Name| Color Code| Usage
Background| #0B1020| Main Background
Surface| #121A2B| Cards
Elevated Surface| #19243A| Dialogs
Primary Accent| #20D6B5| Main Actions
Secondary Accent| #3987FF| Secondary Actions
Primary Text| #FFFFFF| Headings
Secondary Text| #AAB5C8| Supporting Text
Border| #26344A| Dividers

Status Colors

Status| Color
Success| #22C55E
Warning| #F59E0B
Error| #EF4444
Information| #3987FF

These colors are provisional design tokens and may be adjusted after visual testing.

3.2 Typography

Use modern, readable typography.

Recommended fonts:

- Inter.
- Roboto.
- System UI Fonts.

Typography hierarchy:

Element| Style
Main Heading| Bold
Section Heading| SemiBold
Body Text| Regular
Supporting Text| Regular
Balance Amount| Bold
Button Label| Medium

3.3 UI Components

Common components:

- Primary Button.
- Secondary Button.
- Text Input.
- Search Field.
- Information Card.
- Balance Card.
- Status Badge.
- Navigation Bar.
- Bottom Navigation.
- Dialog.
- Confirmation Modal.
- Loading Indicator.
- Error Message.
- Transaction List Item.

All components must use consistent spacing, typography, and corner radius.

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4. ANDROID WALLET UI

Platform: Android.

Recommended Technology: Kotlin + Jetpack Compose.

Design Style:

Modern Dark Cryptocurrency Wallet.

4.1 Splash Screen

Elements:

- SAECOIN Logo.
- SAECOIN Name.
- Initialization Indicator.

The splash screen must be minimal.

No unnecessary advertisements or promotional content.

4.2 Welcome Screen

Screen Elements:

- SAECOIN Logo.
- Welcome Heading.
- Short Description.
- Create Wallet Button.
- Import Wallet Button.
- Security Information Link.

Layout:

Logo
↓
Welcome Message
↓
Create Wallet
↓
Import Wallet
↓
Security Information

4.3 Create Wallet Screen

Elements:

- Screen Heading.

- Security Explanation.
- Generate Wallet Button.
- Loading Indicator.
- Wallet Creation Status.

Recovery phrase screens must use a secure presentation.

The application must warn users not to share their recovery phrase.

4.4 Wallet Dashboard

The Dashboard is the primary wallet interface.

Header

Display:

- SAECOIN Logo.
- Network Indicator.
- Settings Icon.

Balance Section

Display:

- Total SAC Balance.
- Available Balance.
- Pending Balance.
- Hide/Show Balance Button.

Main Actions

Two primary actions:

- Send.
- Receive.

Additional actions:

- Transaction History.
- Addresses.

Network Information

Display:

- Network Connection.
- Current Block Height.

- Synchronization Status.

Recent Transactions

Display:

- Transaction Type.
- Amount.
- Date.
- Confirmation Status.

Bottom Navigation

Navigation Items:

1. Home.
2. Transactions.
3. Addresses.
4. Settings.

4.5 Send Screen

UI Elements:

- Recipient Address Input.
- Paste Address Button.
- QR Scanner.
- SAC Amount Input.
- Available Balance.
- Estimated Fee.
- Total Amount.
- Review Transaction Button.

Transaction Review Screen:

- Recipient Address.
- Amount.
- Transaction Fee.
- Total.
- Confirm Button.

4.6 Receive Screen

UI Elements:

- Wallet Address.
- QR Code.
- Copy Address.

- Share Address.
- Incoming Transaction Status.

The QR code must encode the correct SAECOIN network address.

4.7 Transaction History Screen

Each transaction card must display:

- Transaction Icon.
- Amount.
- Date.
- Transaction Type.
- Confirmation Count.

Transaction Status:

- Pending.
- Confirmed.
- Failed.

Selecting a transaction opens the Transaction Details Screen.

4.8 Transaction Details Screen

Display:

- Transaction ID.
- Inputs.
- Outputs.
- Amount.
- Fee.
- Block Height.
- Timestamp.
- Confirmations.

Include Copy Transaction ID and Open in Explorer actions.

4.9 Wallet Settings Screen

Sections:

Security:

- Change Password.
- Biometric Authentication.
- Backup Wallet.

Network:

- Mainnet/Testnet.
- Node Connection.
- Network Status.

Application:

- Language.
- Notifications.
- Currency Display.
- About SAECOIN.

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5. WINDOWS WALLET UI

Platform: Windows 10/11.

Design Style:

Professional Desktop Cryptocurrency Wallet.

5.1 Main Layout

Use a sidebar-based layout.

Sidebar Navigation

- Dashboard.
- Send.
- Receive.
- Transactions.
- Addresses.
- Network.
- Settings.

Main Content Area

The main content area changes according to the selected navigation item.

5.2 Dashboard

Display:

- Total SAC Balance.
- Available Balance.
- Pending Balance.

- Recent Transactions.
- Network Status.
- Current Block Height.

Quick Actions:

- Send SAC.
- Receive SAC.
- View Transactions.

5.3 Send Screen

Fields:

- Recipient Address.
- Amount.
- Transaction Fee.

Actions:

- Review.
- Confirm.
- Cancel.

5.4 Receive Screen

Display:

- QR Code.
- Public Address.
- Copy Address.
- Generate New Address.

5.5 Network Screen

Display:

- Connected Node.
- Peer Count.
- Block Height.
- Synchronization Status.
- Network Difficulty.

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6. DESKTOP NODE APPLICATION UI

Platform:

- Windows.
- Linux.

Design Style:

Professional Technical Dashboard.

6.1 Main Window

Navigation:

- Overview.
- Network.
- Blockchain.
- Peers.
- Logs.
- Settings.

6.2 Overview Dashboard

Display:

- Node Status.
- Blockchain Height.
- Network Height.
- Sync Progress.
- Connected Peers.
- Data Storage Usage.
- Network Difficulty.

Primary Controls:

- Start Node.
- Stop Node.
- Restart Node.

6.3 Network Screen

Display:

- Connected Peers.
- Peer IP Information where appropriate.
- Connection Duration.
- Peer Protocol Version.
- Connection Status.

Sensitive local network information must not be unnecessarily exposed.

6.4 Blockchain Screen

Display:

- Current Block Height.
- Latest Blocks.
- Block Hash.
- Block Timestamp.
- Block Validation Status.

6.5 Logs Screen

Display:

- Node Startup Events.
- Peer Connections.
- Synchronization Events.
- Block Validation Events.
- Error Messages.

Include:

- Search Logs.
- Filter Logs.
- Export Logs.

6.6 Settings Screen

Options:

- Data Directory.
- Network Selection.
- Peer Configuration.
- Connection Settings.
- Storage Settings.
- Logging Level.

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7. MINING SOFTWARE UI

Platform:

- Windows.
- Linux.

Design Style:

Modern Mining Dashboard.

7.1 Mining Dashboard

Header

Display:

- SAECOIN Logo.
- Network Status.
- Miner Status.

Main Statistics

Display:

- Total Hashrate.
- CPU Hashrate.
- GPU Hashrate.
- Mining Uptime.
- Current Block Height.
- Network Difficulty.

Mining Controls

Buttons:

- Start Mining.
- Stop Mining.
- Pause Mining.

Hardware Section

Display detected hardware:

CPU:

- Processor Name.
- Thread Count.
- Current Usage.
- Hashrate.

GPU:

- GPU Name.
- Device Status.
- Hashrate.

- Temperature (if available).
- Power Usage (if available).

Mining Configuration

Options:

- Solo Mining.
- Pool Mining.
- Wallet Address.
- CPU Thread Configuration.
- GPU Device Selection.

Mining History

Display:

- Mining Sessions.
- Duration.
- Average Hashrate.
- Accepted Shares.
- Rejected Shares.
- Valid Blocks Found.

Estimated rewards must be clearly labeled as estimates.

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8. BLOCK EXPLORER UI

Platform: Responsive Web Application.

Design Style:

Public Cryptocurrency Explorer.

8.1 Homepage

Header

Display:

- SAECOIN Logo.
- Home.
- Blockchain.
- Transactions.
- Network Statistics.

Search Bar

Placeholder:

Search Block Height / Transaction ID / Wallet Address

Network Statistics Cards

Display:

- Current Block Height.
- Total Supply.
- Circulating Supply.
- Network Hashrate.
- Mining Difficulty.
- Total Transactions.

Latest Blocks

Table Columns:

- Height.
- Block Hash.
- Timestamp.
- Transaction Count.
- Block Reward.

Latest Transactions

Table Columns:

- Transaction ID.
- Amount.
- Timestamp.
- Confirmation Status.

8.2 Block Details Page

Display:

- Block Height.
- Block Hash.
- Previous Block Hash.
- Merkle Root.
- Timestamp.
- Difficulty.
- Nonce.
- Block Reward.

- Transaction Count.

8.3 Transaction Details Page

Display:

- Transaction ID.
- Inputs.
- Outputs.
- Transaction Fee.
- Block Height.
- Timestamp.
- Confirmations.

8.4 Address Details Page

Display:

- Address.
- Confirmed Balance.
- Transaction Count.
- Received Amount.
- Sent Amount.
- Transaction History.

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9. RESPONSIVE DESIGN

The Block Explorer must support:

- Desktop.
- Laptop.
- Tablet.
- Mobile Browser.

The layout must automatically adapt to screen size.

Tables may become horizontally scrollable or use compact mobile layouts.

Android Wallet must support different Android screen sizes.

Windows applications must support common desktop resolutions.

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10. ACCESSIBILITY

All applications should support:

- Readable Text.
- Appropriate Color Contrast.
- Clear Button Labels.
- Keyboard Navigation where applicable.
- Screen Reader Labels.
- Visible Focus Indicators.
- Error Messages that explain corrective actions.

Important information must not rely on color alone.

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11. UI PERFORMANCE

The applications must prioritize responsiveness.

Requirements:

- Avoid unnecessary animations.
- Use efficient list rendering.
- Load transaction history progressively.
- Avoid blocking the main UI thread.
- Display loading states during network operations.
- Handle network disconnections gracefully.

Blockchain synchronization must run independently from UI rendering.

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12. SECURITY-RELATED UI RULES

The interface must:

1. Warn users before sending transactions.
2. Display the full recipient address during confirmation.
3. Warn users before exporting private keys.
4. Clearly identify Mainnet and Testnet.
5. Never display recovery phrases in ordinary dashboard screens.
6. Never request private keys from support staff.
7. Clearly identify pending transactions.
8. Show network synchronization status.
9. Warn users when node information is outdated.
10. Prevent accidental duplicate transaction submissions.

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13. DESIGN CONSISTENCY

All SAECOIN applications must follow shared design principles.

Common elements:

- SAECOIN Logo.
- Typography.
- Color Tokens.
- Button Styles.
- Status Indicators.
- Icons.
- Spacing.
- Error Presentation.

Platform-specific layouts are allowed.

The Android Wallet, Windows Wallet, Node Application, Miner, and Explorer do not need identical layouts, but they must share the same visual identity.

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14. FINAL UI ACCEPTANCE CRITERIA

The UI implementation will be accepted when:

1. All required screens exist.
2. Navigation works correctly.
3. All primary buttons perform their intended actions.
4. Loading states are implemented.
5. Error states are implemented.
6. Network status is visible.
7. Mainnet and Testnet are distinguishable.
8. Wallet balances are displayed accurately.
9. Transaction information is correctly displayed.
10. Desktop layouts work at common resolutions.
11. Mobile layouts adapt to different screen sizes.
12. Private information is not unnecessarily exposed.
13. UI does not block blockchain operations.

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15. IMPLEMENTATION RULE

Developers and AI coding agents must follow:

- PRD.md
- AppFlow.md

- UI_Spec.md

The UI must never simulate successful blockchain operations.

All displayed balances, transactions, mining statistics, and network information must originate from actual application state or verified node responses.

Mock data may only be used in explicitly labeled development and UI testing environments.

END OF DOCUMENT